# Virtual Team Boundary Complexity:
# A Team Coordination Perspective

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## Background

This symposium paper was prepared as part of a larger study in which we investigate the concept of "ambidexterity"—defined as the simultaneous rigor and flexibility in project processes—and how it relates to global information system (IS) project success. We proposed a model in a conference article (Lee, DeLone et al. 2007) in which we suggest that the effects of rigor and flexibility are moderated by system requirements dynamism and virtual team boundary complexity. In subsequent refinements of our model we argue that the direct and interactive effects of these variables on IS project success are mediated by team coordination. A key independent variable in this study is "virtual team boundary complexity" and we argue that it is the complexity of the collaboration context that affects coordination, more than the boundaries per se. We have concluded from prior interview studies (Espinosa, Lee et al. 2006) that the coordinative complexity brought about by the presence of multiple boundaries creates substantial collaboration barriers. This is so because team members need to process more information cues to carry out the task, thus the importance of understanding and conceptualizing this contextual complexity. We are presently collecting data for this study and we are using a preliminary operationalization of this boundary complexity construct, which we have developed using coordination theory (Van de Ven, Delbecq et al. 1976; Malone and Crowston 1990; Malone and Crowston 1994), complexity literature (Wood 1986), and virtual teams research (Orlikowski 2002; Watson-Manheim, Chudoba et al. 2002; Espinosa, Cummings et al. 2003; Griffith, Sawyer et al. 2003; Kirkman and Mathieu 2005; Lu, Watson-Manheim et al. 2006). We hope that the symposium discussion will help us further develop this important construct.

**Introduction**

Given current globalization and outsourcing/offshoring trends, many technical projects contexts can be characterized as very virtual and global. Technical projects increasingly employ virtual teams working in complex, geographically distributed collaboration arrangements as organizations seek to develop and implement effective systems for users and customers around the world at lower costs by leveraging internal and external resources (Powell, Piccoli et al. 2004). These virtual teams are divided by various boundaries, including distance, temporal, organizational and cultural (Lipnack and Stamps 1997; Chudoba, Crowston et al. 2002; Orlikowski 2002; Espinosa, Cummings et al. 2003). These boundaries often co-exist within a given team and the resulting complexity of collaborating across these multiple boundaries becomes an important risk factor for project success.

The literature on virtual teams has started to address the concept virtuality or virtualness in teams (Griffith, Sawyer et al. 2003; Kirkman and Mathieu 2005; Lu, Watson-Manheim et al. 2006). While this is a step in the right direction, these constructs have limitations in helping us study how teams coordinate their work across complex geographic configurations in teams. Some examples of unanswered questions include: how does virtualness change as the number of locations or time zones changes in a team or as the distribution of team members across these locations or time zones varies? Or, more importantly, how does this affect how teams coordinate their work? While the concept of virtualness is important to characterize and study virtual teams, we suggest that the complexity of the virtual team boundary context helps us better understand the coordination challenges faced by virtual teams.

In this research we develop and propose a construct for virtual team boundary complexity that can help us understand how teams can coordinate their work more effectively across these boundaries and suggest some measures to measure this complexity. Coordination is about managing dependencies among task activities (Malone and Crowston 1994). We posit that it is not the virtual team boundaries themselves that makes it more difficult to manage these dependencies, but the complexity of the virtual team boundary configuration resulting from having to work across multiple boundaries. Therefore, understanding this complexity is very important in figuring out how virtual teams can coordinate their work more effectively. We draw upon data collected by the authors and colleagues for two different research projects (Espinosa, Lee et al. 2006; Espinosa and Pickering 2006) to support our arguments.

**Virtual Team Boundary Complexity and Coordination Effectiveness**

Virtual team boundaries create barriers that make it difficult for members to communicate and work together (Orlikowski 2002; Watson-Manheim, Chudoba et al. 2002; Espinosa, Cummings et al. 2003; Cummings 2004). How these various boundaries are arranged within a team can make a big difference. Take for example the familiar outsourcing relationship between a U.S. and an Indian firm working together on a technical project. Team members in this context are in only two locations, but the geographical boundary dividing this team aligns perfectly with organizational, distance, time zone, cultural and language boundaries. This alignment of boundaries creates a “fault line,” which has been argued to be very difficult to bridge (Lau and Murnighan 1998). However, one of our studies (Espinosa, Lee et al. 2006) found evidence that team members in two such locations adjusted quite well and learned how to operate effectively with each other because the location, time, organizational, cultural and language differences were well understood by all, such that team members were able to implement effective coping mechanisms (e.g., shift work hours, more detailed documentation, use of mobile communication devices, etc.) to work together. In contrast, the other study (Espinosa and Pickering 2006) found that when a team operates in several locations spanning multiple time zones, cultures, languages, and organizations things become more unpredictable and the team has a more difficult time finding a rhythm to coordinate their work effectively. In fact, there were so many locations and time zones represented in one team we studied that team members shifted their work hours so dramatically that some co-located team member had non-overlapping work hours. We refer to this diversity of boundaries as “virtual team boundary complexity.”

Wood (1986) argued that task complexity increases when more information cues need to be processed to carry out the task and indicated that tasks not only have a “structural” component that is inherent to the task, but also have a “coordinative” complexity that is affected by the task context (Xia and Lee 2005). For example, when a task is carried out by many team members their actions need to be coordinated, adding further complexity to the execution of the task. When the context makes it difficult to exchange the information cues necessary to carry out the task – i.e., the task context is more complex – it affects the team’s ability to coordinate the task, but this complexity is not captured in the concept of virtualness. The virtualness dimensions proposed in the literature include: physical distance among team members; level of technology support; percentage of time apart in the task (Griffith, Sawyer et al. 2003); synchronicity – to distinguish between “real-time” and “lagged-time” interaction (Kirkman and Mathieu 2005); and workplace mobility – to capture the extent to which team members work at various sites, telecommute and use mobile devices (Lu, Watson-Manheim et al. 2006). These dimensions are useful but have limitations in helping us understand coordination, particularly when teams are dispersed in more complex geographic configuration arrangements involving several sites. For example, some teams may be widely dispersed (i.e., balanced), while others may have the majority of members in one site with a few isolated members in other sites (i.e., unbalanced) (O'Leary and Mortensen 2005), and some teams may be dispersed on a North-South axis with little time zone difference, while others may be dispersed along an East-West axis with substantial time zone differences (O'Leary and Cummings 2002; Espinosa and Pickering 2006).

We argue that coordination is affected when virtual team boundary arrangements make it difficult to communicate – i.e., coordinate “organically” – because the team is forced to coordinate “mechanistically” using task programming mechanisms like schedules, plans, and division of labor, which are not as effective when task conditions are less routine (March and Simon 1958). While each type of boundary will have its unique effects on communication, we argue that the overall communication effectiveness in the team and, therefore, its ability to coordinate organically will be generally affected by four dimensions. : (1) the different of types of boundaries spanned by the team; (2) the average number of boundaries present for each boundary type; (3) the extent to which these boundaries align; and (4) the dispersion of members across these boundaries. Consequently, we propose that these dimensions define a virtual team's boundary complexity. We briefly elaborate on each of these.

Task complexity increases when the number of information cues that need to be processed increases (Wood 1986). This complexity may be due to structural characteristics of the task or due to the characteristics of the task context. In this paper we focus on the characteristics of the task's virtual context.

Members in teams that span more boundary types (e.g., distance, time zones, organizational, cultural) need to process more information cues to get the job done when there are more boundaries represented in the team (Espinosa, Cummings et al. 2003). For example, members of a team without virtual boundaries (e.g., co-located, same work hours, same culture, same organization) will become very familiar very quickly with their task context and each others' work habits. Similarly, a team with only one type of virtual boundary (e.g., different locations, same time zones, same culture, same organizations) will have to face the added complexity of having to process information cues associated with other locations and distance (e.g., figuring out when colleagues are around, knowing who is who in the other location, figuring out what is the best way to communicate with someone, etc.). Therefore, compared to co-located teams, the complexity of the virtual collaboration context will increase when distance boundaries are represented in a team, thus making it more difficult to coordinate. Similarly, the complexity of the virtual collaboration context will increase as more boundary types (e.g., more than one time zone, more than one culture, etc.) are represented in a team.

> ***Proposition 1:*** *Virtual team boundary complexity increases when the number of boundary types represented in a team increases.*

Similarly, team members will also have more difficulty communicating and coordinating when there are more boundaries within the various boundary types. For example, it is more difficult to coordinate work when teams operate across several locations (O'Leary and Cummings 2002) or across multiple time zones (Espinosa and Pickering 2006) than when the there are only two or three locations or time zones. For example, a team with a distance boundary operating in only two locations will develop familiarity about the work habits and other idiosyncrasies of both locations much faster than a team operating in multiple locations. As more locations are involved, team members need to process more information cues specific to each location. Similarly, teams with more than two time zones, cultures and organizational memberships will need to process more information cues than teams with only two time zones, cultures and organizational memberships, respectively.

> ***Proposition 2:*** *Virtual team boundary complexity increases when the number of boundaries spanned by the team for each boundary type increases.*

Boundary alignment can also affect communication and coordination. Lau and Murnighan (1998) suggested that it is more difficult to work together when group attributes correlate within a subgroup rather than cutting across subgroups because this creates "fault lines", which exacerbate team dynamics with subgroups and fracture a team into subgroups. In the context of geographically distributed teams, fault lines occur when team boundaries correlate within one geographic location more strongly than across geographic locations, resulting in a stronger salience of subgroups by location (Cramton and Hinds 2005). Naturally, one would expect fault lines across geographic locations in which multiple boundaries align to hinder coordination. However, data from our prior studies suggest that when teams have incentives to produce and pressures to deliver, team members adjust and learn how to bridge these fault lines (Espinosa, Lee et al. 2006), whereas multiple misaligned boundaries create more confusion about how and when to interact (Espinosa and Pickering 2006). We argue that this confusion stems from the fact that team members need to process more information cues – i.e., the collaboration context is more complex – to figure out who to communicate with and how to interact with them. A team that is divided by a given number of boundary types and a given number of boundaries within each type will need to process more information cues when the different boundaries do not align. For example, a team with distance, time zone, cultural and organizational boundaries operating in three sites will develop familiarity about their collaboration context faster if each location has the same work hours, culture and organizational membership. Despite the fact that sites are divided by fault lines because all the boundaries align, members will figure out how to collaborate with each site. Conversely, if the same boundary types are distributed across locations, such that each location has more than one set of work hours (i.e., time separation), cultures and organizational memberships, team members will need to process more information cues to collaborate with each of the other sites.

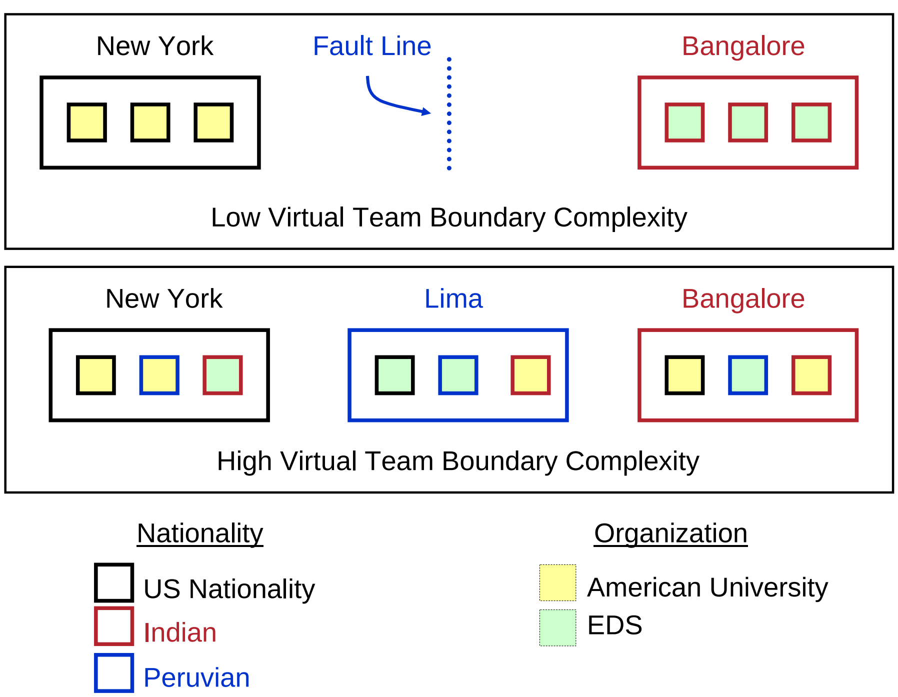
> ***Proposition 3:*** *Virtual team boundary complexity increases when boundary alignment decreases.*

Finally, recent research has provided evidence that a team's geographic configuration – i.e., how team members are distributed across locations, makes a difference (Espinosa, Cummings et al. 2006; O'Leary and Cummings 2007) on the collaboration context. For example, one team with 12 members operating in 3 sites within the same time zone and without cultural or organizational membership differences will need to process more information cues if there are 4 members in each site (i.e., widely dispersed or evenly balanced) than if 10 members are in one central location and each of the other two members is in each of the other two locations (i.e., unevenly balanced or unbalanced). The same is true with respect to time zones, cultures and organizational memberships. That is, team members will need to process more information cues when members are widely spread (or evenly balanced) distribution across time zones, cultures and organizations, than when team members are largely concentrated within a given time zone, culture and organizational affiliation respectively. For example, a team with a high concentration of members in the US and one or two isolated members in Indian less difficulty working as a group than if team members are equally divided between the two locations, who will need to bridge more boundaries to communicate and coordinate.

> ***Proposition 4:*** *Virtual team boundary complexity increases when members are more widely dispersed across boundaries.*

### Illustration

The figure below illustrates how as the number of boundary types, number of boundaries, misalignment of boundaries and dispersion of members across boundaries increases, the complexity of the collaboration context increases too. The result is that team members need to process more information cues to carry out the task because there are more boundary factors (language, local times, presence awareness of colleagues, etc.) to consider. At the same time, this boundary complexity makes it more difficult to provide and acquire these information cues, precisely when they are needed the most, thus severely affecting the team's ability to coordinate.



## Measurement

We are in the process of collecting data to validate this construct to be used as a key independent variable affecting coordination in a related study. One of the challenges of this research is developing valid measures for this construct. We have developed preliminary measures for large teams in which there is one respondent per project, who is providing responses on behalf of the team. Naturally, some granularity in the measure is lost with this approach but we see no other practical solution when development teams are quite large. In our current study, we are collecting data on 4 boundary types: distance, time zones, cultural and organizational. We ask participants to provide a member count for each project location, nationality and organization involved in the project, as illustrated in the three screenshots from the actual web instrument we are using. We then measure the respective construct items as follows:

1. Number of boundary types: can be computed by counting how many boundary types are represented in the team. In the example below we are considering 4 types of boundaries: distance, time zone, culture and organizational membership. All 4 boundary types are represented in this team so this variable measure is 4.
2. Average number of boundaries for each type: can also be computed from an average of the number of boundaries for each boundary type. In the example below there are 3 locations, 2 time zones, 5 cultures and 2 organizations, so the average value for this variable is 2.5.
3. Boundary alignment: this item cannot be computed from data for large teams collected this way. We can compute some proxy approximate measure by applying some correlation or similarity metrics to evaluate how the numbers provided for boundary align with each other.
4. Dispersion of members across these boundaries: this item can be computed using variance or homogeneity measures. Using Gini coefficients (Dorfman 1979) to measure homogeneity (i.e., 0 = fully concentrated within 1 boundary; 1 = evenly distributed across all boundaries)

we obtain values of 0.64, 0.70, 0.71 and 0.77 for distance, time zones, cultures and organizational memberships respectively, with an average of 0.705 across all 4 boundaries.

### Geographic Distribution of Team Members

**Page no.4**

In how many cities did this project team have members (enter a number)?

3

Submit Reset

| # | City | State or Province | Country | Number of Team Members (please enter a number) |
|---|---|---|---|---|
| 1 | Washington | DC | USA | 10 |
| 2 | Lima | Lima | Peru | 2 |
| 3 | Seoul | Seoul | Korea | 3 |
| To delete a row enter 0 team members | | | | |

Submit Reset

### Cultural and National Origin of Team Members

**Page no.5**

How many nationalities were represented in the team (enter a number)?

5

Submit Reset

| # | Country of Origin | Number of Team Members (please enter a number) |
|---|---|---|
| 1 | USA | 6 |
| 2 | India | 3 |
| 3 | Israel | 1 |
| 4 | Peru | 2 |
| 5 | Korea | 3 |
| To delete a row enter 0 team members | | |

Submit Reset

### Number of Organizations Involved in the Project

**Page no.6**

How many different organizations were represented in the team (enter a number)?

2

Submit Reset

| # | Organization | Number of Team Members (please enter a number) |
|---|---|---|
| 1 | American University | 11 |
| 2 | EDS | 4 |
| To delete a row enter 0 team members | | |

Submit Reset

For smaller teams, the measures can be developed more precisely. We only suggest guidelines for these measures because we have not collected data for small teams. In smaller teams it is more feasible to include all team members in the response sample. Because a specific boundary separates any two individuals in the team, social network analysis methods are ideally suited to measure the relational properties of the boundaries that separate each dyad in the team. Therefore, data can be collected at the dyad level, as is typical in social network studies. First, the number of boundaries type (e.g., distance, time zones, etc.) present in each team can be counted for the team as a whole. Then, a sociomatrix can be created for each boundary type and the following measures computed for each dyad:

1. Average number of boundaries for each boundary type: enter 1 in sociomatrix if the dyad is divided by the boundary, 0 otherwise; then compute the density of the sociomatrix, relative to the maximum number of boundaries possible.
2. Boundary alignment: use similarity metrics among sociomatrices for the various boundary types.
3. Dispersion of members across these boundaries: use variance or homogeneity measures

Naturally our proposed construct needs further refinement and our proposed measures need to be validated. However, we believe that this is a step in the right direction and, as a community of virtual team researchers, we need to better understand how the complexity of the collaboration context affects the team's ability to get the work done efficiently and effectively. He hope to be contributing towards this end, but also look forward to our symposium discussion shedding some light into better approaches.

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